Tracking Control of Inverted Pendulum on a Cart with Disturbance using Pole Placement and LQR

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***Abstract*- This paper presents the Mathematical non-linear model of Inverted Pendulum (IP) system on a cart based on Lagrange's Equation. The linearization is done through Tayler's series approximation. Linear Quadratic Regulator (LQR) and Pole Placement control strategy are used for solving tracking problem and stabilization of pendulum around upright position. Square wave is used as reference signal for IP tracking. Simulation results are compared with and without disturbances for both the control strategy. The performance of controllers is obtained based on specification of response time and disturbance rejection level. The performance of LQR is better than pole placement control.**

Keywords- Inverted Pendulum, LQR, Pole Placement control, Tracking problem, Stabilization.

I. INTRODUCTION

The inverted pendulum(IP) is an open loop unstable system with high non-linear dynamics. IP system modelling and control provide solution to many real-time problems which make it pivotal area of research in control system [1,2]. Broadly two main challenges faced by control engineer concerning IP system are Pendulum stabilization about upright position and cart position tracking problem [3,4].

In literature, for angle stabilization Proportional Derivative Integral(PID) controller were used [4] but it does not provide the desired results. In [5]; Multiple PIDs were used to control angle and Servo problem simultaneously. In [6]; the tracking as well as stabilization control of IP system with actual nonlinear model using PID controllers was investigated but choosing the controllers parameters was not clarified. In [8]; fractional PID tuned by particle swarm optimization(PSO) is discussed. Cascaded PID controller using Genetic Algorithm is shown in [7]. In Linear Quadratic Regulator(LQR) control strategy weighting matrices i.e. Q, R are assumed to obtain optimal condition for states and control input[9-11]. The system with disturbances were improved using Linear Quadratic Gaussian(LQG) control[10,12]. In [4]; LQR and PID controller performance were independently obtained with and without disturbance rejection.

This paper presents the design of pole placement control, LQR control for angle stabilization and tracking cart position. Results are investigated with and without disturbances. The performance of controller is discussed based on time response specification and level of disturbance rejection.

There are six sections organized in this paper. Section I features introduction and literature review. IP System description, modelling (non-linear) linearization about upright position (equilibrium position) and state space model was derived in section II. Tracking problem formulation and different control strategy were discussed in section III. MATLAB simulation of IP system was shown in section IV. In section V; results and discussion were done. At the end, concluding statement were offered in section VI.

II. INVERTED PENDULUM

In Fig.1 IP system on cart; which can move along X-axis horizontally. One end of the pendulum rod is fixed at pivot point while the other end is free to move in vertical X-Y plane. When the pendulum at upright position it is called IP system. The IP system is unstable hence it requires continuous balancing force F supplied by Motor to keep the pendulum upright.

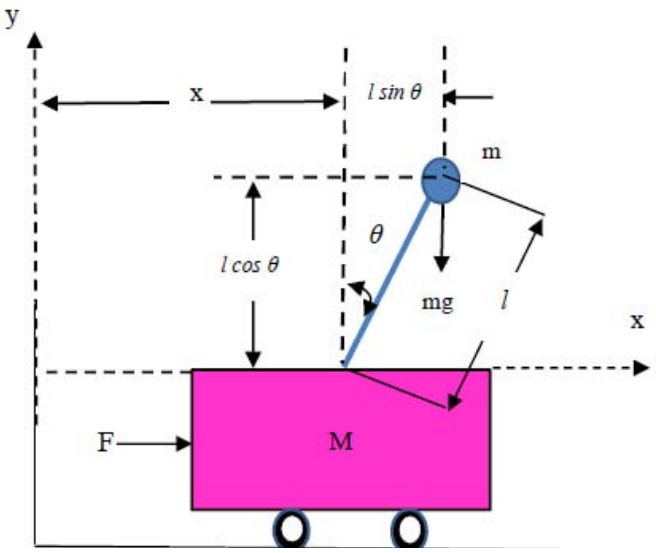


Fig. 1 The inverted pendulum system[14]

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A. Non-linear System Modelling

In this section, non-linear mathematical model of IP cart system is derived. Few assumptions were made for simplicity:

- Mass of the rod is zero.
- Friction coefficient between rod and cart is neglected.

Kinetic energy of cart

$$KE_c = \frac{1}{2}M\dot{x}^2 \quad (1)$$

In the Fig.1 co-ordinate position of the pendulum

$$X_p = x + l\ \sin\theta \quad (2)$$

$$Y_p = l\ \cos\theta \quad (3)$$

Differentiating (2) & (3) results in

$$V_{px} = \dot{x} + \dot{\theta}l\ \cos\theta \quad (4)$$

$$V_{py} = -\dot{\theta}l\ \sin\theta \quad (5)$$

Velocity of pendulum using (4) & (5)

$$\left|V_p\right|^2 = {V_{px}}^2 + {V_{py}}^2 \quad (6)$$

Kinetic energy of pendulum using (6)

$$KE_p = \frac{1}{2}m\left[\dot{x}^2 + 2l\dot{\theta}\dot{x}\cos\theta + \dot{x}^2l^2\right] \quad (7)$$

Kinetic energy of the system by adding (1) & (7)

$$KE = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left[\dot{x}^2 + 2l\dot{\theta}\dot{x}\cos\theta + \dot{x}^2l^2\right] \quad (8)$$

Potential energy of pendulum

$$T = -mgl\cos\theta \quad (9)$$

Mathematical model of the inverted can easily be derived by using Lagrange's equation[3,8]

$$L = KE - T \quad (10)$$

Choosing x, θ as two coordinates describing the system

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = q_x \quad (11)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = q_\theta \quad (12)$$

Where,

$$q_x = F \qquad q_\theta = 0 \quad (13)$$

Substituting (8), (9) & (10) in (11) & (12) results in

$$(M+m)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta = F - \mu x \quad (14)$$

$$ml\ddot{x}\cos\theta - ml^2\ddot{\theta} + mgl\sin\theta = 0 \quad (15)$$

B. Linearization of IP System

The linearization of non-linear IP system about upright position i.e. $\theta = 0$ using Tayler's series approximation.

$$f(\gamma) = \sum_{n=0}^{\infty} \frac{(\gamma-\gamma_0)^n}{n!} f^n(\gamma_0) \quad (16)$$

Where, γ_0 is the equilibrium point around which system is linearized.

Assuming,

$$\cos\theta \cong 1 \qquad \sin\theta \cong \theta \qquad \dot{\theta}^2 = \dot{\emptyset}^2 \cong 0 \quad (17)$$

Using (14), (15) & (17) linearized equations of the system obtained in (18) & (19)

TABLE I. PARAMETERS OF THE IP SYSTEM [13]

Parameter	Value	Unit
Mass of the Cart 'M'	2.4	kg
Mass of the Pendulum 'm'	0.23	kg
Length of the rod 'l'	0.36	m
Friction Coefficient between the Cart & base 'μ' (assumed)	0.1	$Nm^{-1}s^{-1}$
Force of Gravity on system 'g'	9.81	m/s^2
Length of the Track 'D'	±0.5	m
Force applied to the cart 'F'	-	N
Position of the cart 'x'	-	m
Pendulum angle 'θ'	-	rad

$$\ddot{x} = \frac{F - \mu\dot{x} + ml\ddot{\theta}}{(m+M)} \quad (18)$$

$$\ddot{\theta} = \frac{-(\ddot{x} + g\theta)}{l} \quad (19)$$

Rearranging (18) & (19) in terms of $\ddot{\theta}$ & $\ddot{x}$ as

$$\ddot{x} = \frac{F - \mu\dot{x} + mg\theta}{M} \quad (20)$$

$$\ddot{\theta} = \frac{-F + \mu\dot{x} - g(m+M)\theta}{Ml} \quad (21)$$

Suppose state variables are

$$x_1 = \theta \qquad x_2 = \dot{\theta} \qquad x_3 = x \qquad x_4 = \dot{x} \quad (22)$$

Let output be

$$y_1 = \theta = x_1 \qquad y_2 = x = x_3 \quad (23)$$

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C. State Space Model

State space model is represented as

$$\dot{x}(t) = Ax(t) + Bu(t) \tag{24}$$

$$y(t) = Cx(t) + Du(t) \tag{25}$$

Using (20)-(25); results in

System matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -g(m+M)/Ml & 0 & 0 & \mu/Ml \\ 0 & 0 & 0 & 1 \\ mg/M & 0 & 0 & -\mu/M \end{bmatrix} \tag{26}$$

Control matrix

$$B = \begin{bmatrix} 0 \\ -1/Ml \\ 0 \\ 1/M \end{bmatrix} \tag{27}$$

Output matrix

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{28}$$

Feed-forward matrix

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{29}$$

III. Controllers

A. Tracking Problem Formulation

In fig. 2, ***r*** denotes reference position of cart which to be tracked by the IP system. The error ***e*** in the tracking problem is eliminated by Integral control action. Hence a new state ***z*** is augmented to plant state vector ***x***.

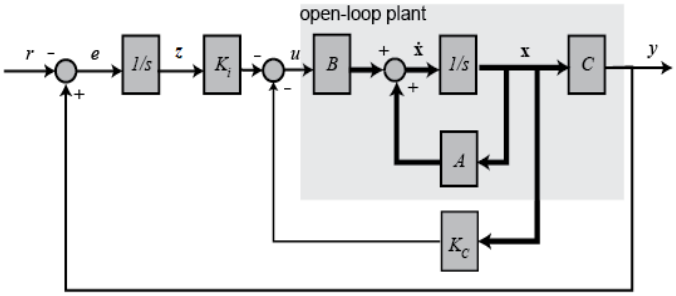


Fig. 2 State Feedback with Integral Control

$$z(t) = \int_0^t (y(t) - r)\, dt \tag{30}$$

Differentiating on both side in (30) gives

$$\dot{z}(t) = y(t) - r = Cx(t) - r \tag{31}$$

Using (24) & (31) augmented system obtained

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ -1 \end{bmatrix} r \tag{32}$$

At steady state

$$\dot{x} = 0 \qquad \dot{z} = 0 \tag{33}$$

Substituting (33) in (32) results

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix} r = -\begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \begin{bmatrix} x_s \\ z_s \end{bmatrix} - \begin{bmatrix} B \\ 0 \end{bmatrix} u \tag{34}$$

Putting (34) in (32)

$$\begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \begin{bmatrix} x - x_s \\ z - z_s \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} [u - u_s] \tag{35}$$

Let,

$$\tilde{x} = \begin{bmatrix} x - x_s \\ z - z_s \end{bmatrix} \qquad \tilde{u} = [u - u_s] \tag{36}$$

Put (36) in (35) results

$$\dot{\tilde{x}} = \bar{A}\tilde{x} + \bar{B}\tilde{u} \tag{37}$$

Where,

$$\bar{A} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \qquad \bar{B} = \begin{bmatrix} B \\ 0 \end{bmatrix} \tag{38}$$

$$\tilde{u} = -k\tilde{x} \tag{39}$$

k can be partitioned as

$$k = [k_p \quad k_i] \tag{40}$$

Where,

k_i is integrator gain

$k_p = [k_1 \quad k_2 \quad \ldots \quad k_n]$ for **n** states system.

System should be completely controllable is prerequisite for implementing pole placement and LQR[5].

$R = [B \quad AB \quad \ldots \quad A^{n-1}B]$

Rank(R) = n is the condition for controllability.

B. Pole Placement Method

In this control strategy closed loop poles are placed in left half of complex s-plane to make open loop unstable system stable and better transient response.

Let set design requirement for the IP system as

$$M_p \leq 10\% \qquad t_s \leq 4\ sec \tag{41}$$

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$$M_p = e^{\frac{-\pi\xi}{\sqrt{1-\xi^2}}} \qquad t_s = \frac{4}{\xi\omega_n} \qquad \omega_d = \omega_n\sqrt{(1-\xi^2)} \quad (42)$$

Substituting (41) in (42) results in

$$\omega_d = 1.364 \qquad \omega_n = 1.69 \qquad \xi = 0.591 \quad (43)$$

Dominant poles using (43) are

$$P_{1,2} = -\xi\omega_n \pm j\,\omega_d = -1 \pm j1.364 \quad (44)$$

As, there are five states in augmented system (40) to incorporate integral control action; three arbitrary poles were placed at least five times away from the location of dominant poles in the left side of s-plane on the real axis.

C. Linear Quadratic Regulator (LQR)

The objective of LQR is to obtain control signal which satisfy given physical constraints of system along with minimizing the cost function ***J***.

$$J = \int_0^\infty (x^T Q x + u^T R u)\, dt \quad (45)$$

Algorithm for solving LQR problem,

Step 1: Initialise state weighting matrix *Q* and control force weighting matrix *R* matrices.

Where, $Q \in R^{n\times n}$ is positive semi-definite, symmetric matrix and ***n*** denote states.

R is positive definite, symmetric matrix.

Step 2: Solve algebraic Riccati equation given below to find Riccati coefficient matrix *P*

$$PA + A^T P + Q - PBR^{-1}B^T P = 0 \quad (46)$$

Step 3: Compute control input gain *K* using P

$$K = R^{-1}B^T P \quad (47)$$

Step 4: Compute optimal control action

$$u(t) = -Kx(t) \quad (48)$$

Step 5: Obtain quadratic performance index *J* as in (45)

IV. Simulation

State space matrices of the IP system on cart was obtained substituting Table 1 parameters in (24)-(29) as

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -29.86 & 0 & 0 & 0.1157 \\ 0 & 0 & 0 & 1 \\ 0.94 & 0 & 0 & -0.0417 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ -1.157 \\ 0 \\ 0.4167 \end{bmatrix} \quad (49)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (50)$$

A. Pole Placement Method

From Eq.48 dominant poles are at $-1 \pm j1.364$ and three arbitrary poles were placed shown below

$$P = [-5 \quad -5.5 \quad -6 \quad -1 + j1.364 \quad -1 - j1.364] \quad (51)$$

Resultant gain matrix

$K = [-64.8272 \quad -4.0846 \quad 51.43 \quad 32.9313 \quad 40.8247]$

Using (40), we get

$K_p = [-64.8272 \quad -4.0846 \quad 51.43 \quad 32.9313]$

$K_i = [40.8247]$

B. Linear Quadratic Regulator (LQR)

Initializing Q and R matrix

$$Q = \begin{bmatrix} 400 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 500 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 650 \end{bmatrix} \qquad R = 1 \quad (52)$$

Results in

$K = [-18.0080 \quad -0.8887 \quad 37.8656 \quad 18.2128 \quad 25.4951]$

Using (40)

$K_p = [-18.0080 \quad -0.8887 \quad 37.8656 \quad 18.2128]$

$K_i = [25.4951]$

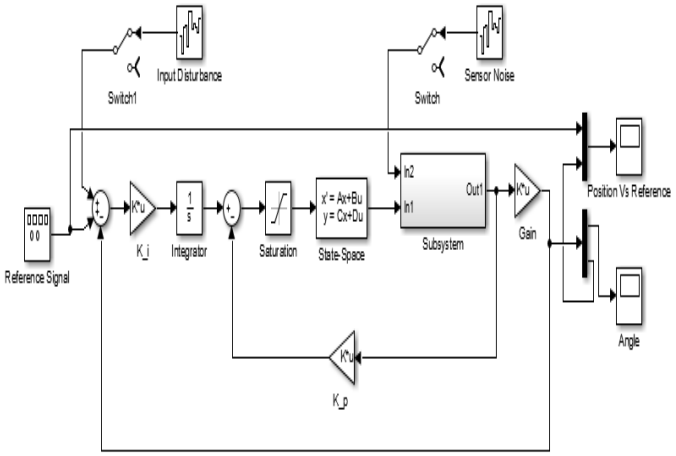


Fig. 3 Pole placement / LQR simulink block

V. Results

In this section; plots of IP system on a cart using Pole placement and LQR strategy were obtained with & without disturbances.

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A. Without Disturbance

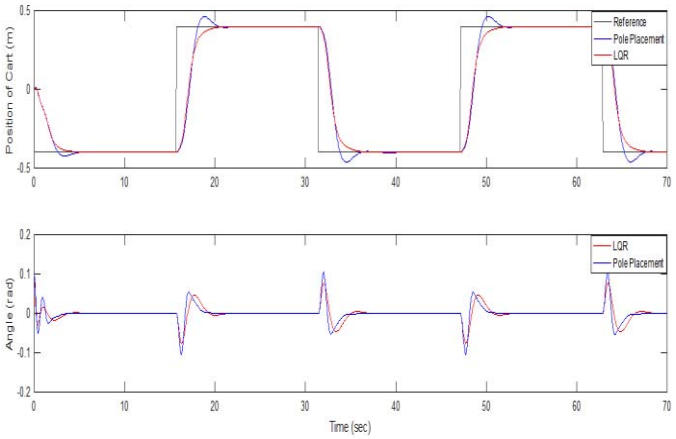


Fig. 4 Tracking response of IP system

B. With Input Disturbance

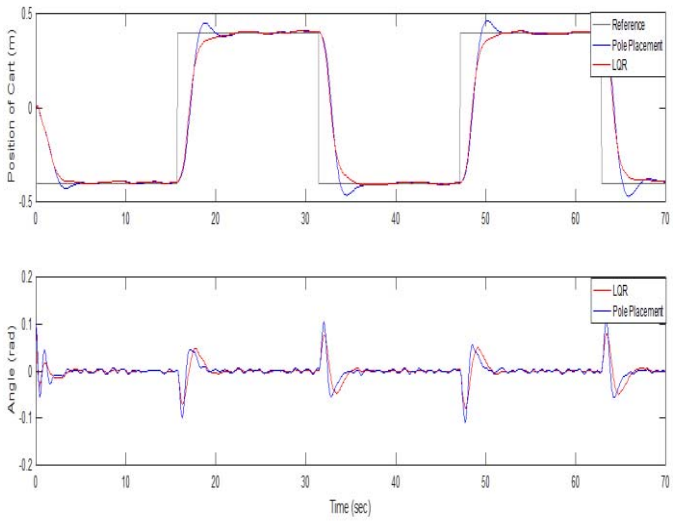


Fig. 5 Tracking response of IP system with input disturbance

C. With Sensor Noise

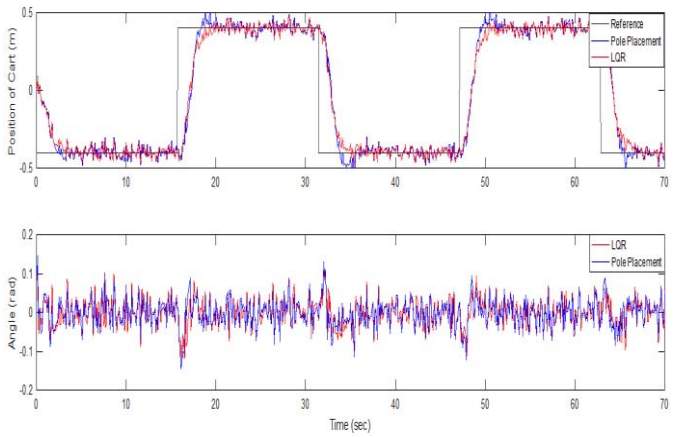


Fig. 6 Tracking response of IP system with sensor noise

TABLE II. SYSTEM RESPONSE WITHOUT DISTURBANCE

Controller	System Parameters		
	M_p (%)	t_s (sec)	$\theta_{max}(rad)$
Pole Placement	12	5.4	0.110
LQR	0	4	0.075

TABLE III. SYSTEM RESPONSE WITH INPUT DISTURBANCE

Controller	System Parameters		
	M_p (%)	t_s (sec)	$\theta_{max}(rad)$
Pole Placement	12.5	11.4	0.12
LQR	1.5	9.2	0.09

TABLE IV. SYSTEM RESPONSE WITH SENSOR NOISE

Controller	System Parameters		
	M_p (%)	t_s (sec)	$\theta_{max}(rad)$
Pole Placement	30	–	0.148
LQR	20	–	0.135

VI. CONCLUSION

In this paper; the examination of tracking problem and stabilization control of IP system was done using pole placement and LQR controllers. The simulation of system with different kinds of noises such as input disturbance and sensor noise were performed. Without disturbance; the IP system tracking the reference signal and stabilization were perfect. With pole placement strategy peak overshoot and settling time exceeded the desired limit as mentioned in (41) but with LQR the system performances were within the desired limit. LQR performed better than pole placement technique with input disturbance and satisfactory results were obtained. With sensor noise substandard results were obtained in either of the controller. A good disturbance rejection was produced for input disturbance but for sensor noise.

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